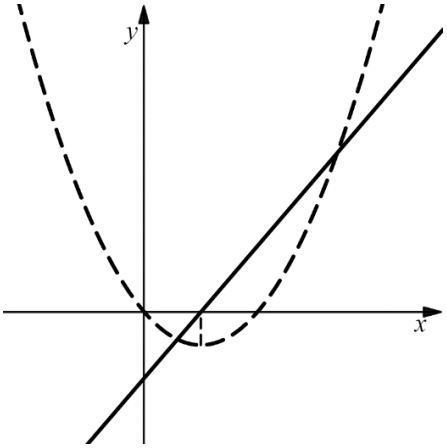
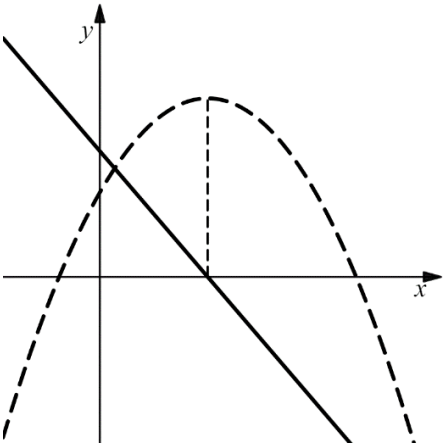
1.33 Differentiation: Graphs of Gradient Functions

Worked Solutions

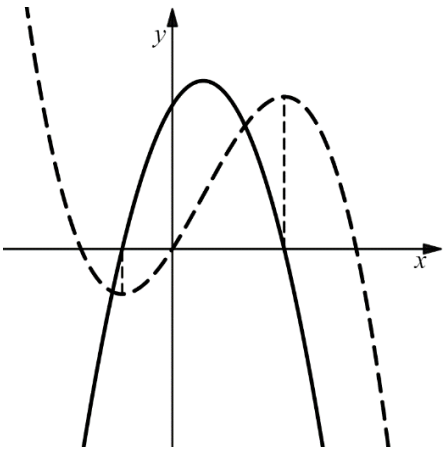
1 a)



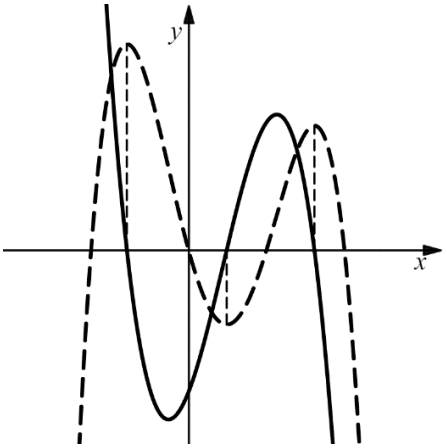
1 b)



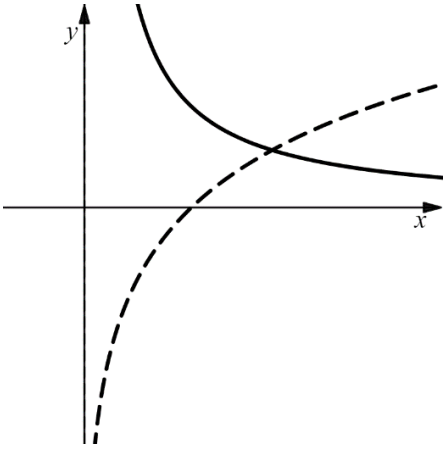
1 c)



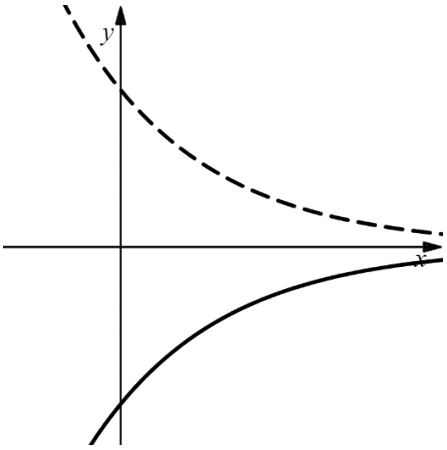
1 d)



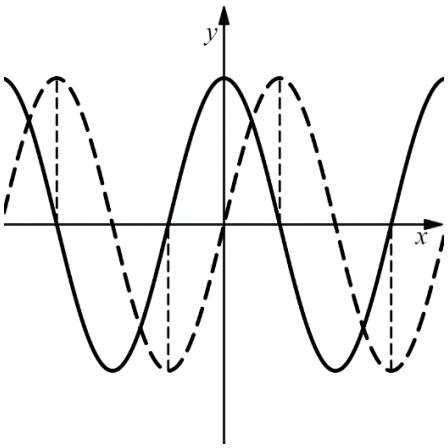
1 e)



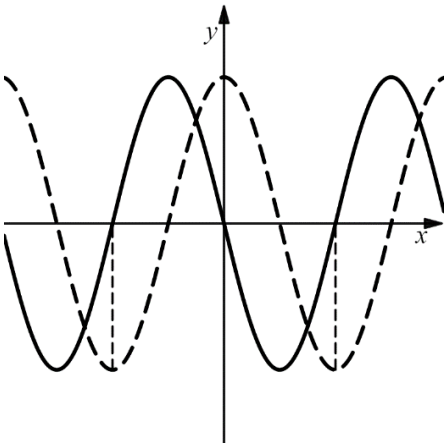
1 f)



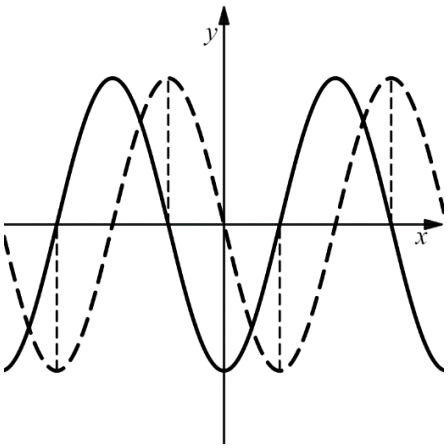
2 a)



2 b)



2 c)



3 a)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

$a = 0$, b is positive, $c = 0$

3 b)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

$a = 0$, b is negative, c is positive.

3 c)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

$a = 0$, b is negative, c is negative.

3 d)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

a is positive, $b = 0$, c is positive.

3 e)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

a is negative, b is positive, $c = 0$.

3 f)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

a is positive, b is positive, c is positive.

3 g)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

a is negative, b is positive, c is negative

3 h)

$$\mathrm{f}'(x) = 3ax^2 + 2bx + c$$

a is positive, b is negative, c is negative

4)

